



NeuroSpace

Complete evaluation export

Job ID	95487eaa-9019-4817-a783-68b64c24a3e8
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Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	20
manual_overrides.qrnn_epochs	10
manual_overrides.batch_size	32
manual_overrides.rnn_batch_size	32
manual_overrides.qrnn_batch_size	16
num_classes	2
class_names	0, 1
qrnn_backend	qiskit
noise_enabled	True
noise_level	0.0010
mitigation_enabled	True
mitigation_runs	3

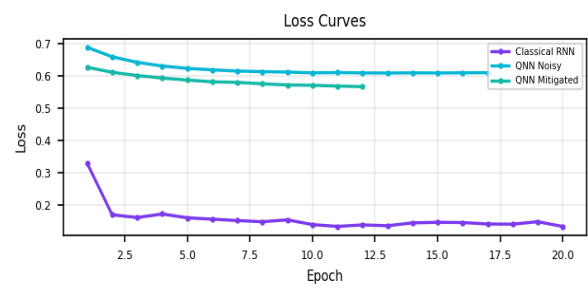
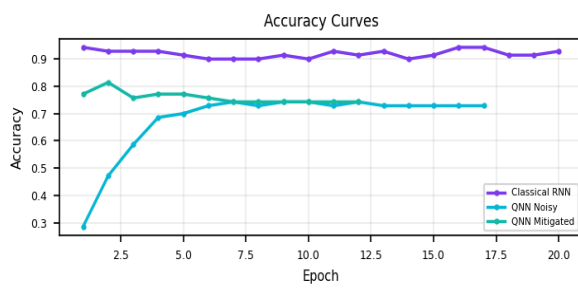
Best Model

Winner	RNN Accuracy	QNN Accuracy	Improvement
RNN	96.43%	77.14%	19.29%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	96.43%	96.57%	96.43%	96.45%	0.1340
QNN Noisy	75.71%	75.06%	75.71%	75.12%	0.6106
QNN Mitigated	77.14%	76.67%	77.14%	76.00%	0.5674

Learning Curves



Classical RNN

Metric	Value
Accuracy	96.43%
Precision	96.57%
Recall	96.43%
F1 Score	96.45%
Loss	0.1340
Train Loss	0.0653
Val Loss	0.1340
Train Accuracy	97.55%
Val Accuracy	92.86%
Train Size	489
Val Size	70
Test Size	140

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	1	47

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
0	98.88%	95.65%	97.24%	92.0000
1	92.16%	97.92%	94.95%	48.0000
macro avg	95.52%	96.78%	96.09%	140.0000
weighted avg	96.57%	96.43%	96.45%	140.0000
Accuracy			96.43%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.4398	0.3298	82.82%	94.29%
2	0.1307	0.1703	96.93%	92.86%
3	0.1139	0.1617	97.14%	92.86%
4	0.0917	0.1727	98.16%	92.86%
5	0.0969	0.1609	96.93%	91.43%
6	0.0742	0.1567	97.55%	90.00%
7	0.0745	0.1524	98.16%	90.00%
8	0.0762	0.1487	97.34%	90.00%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.0760	0.1546	97.75%	91.43%
10	0.0679	0.1399	98.16%	90.00%
11	0.0777	0.1342	98.16%	92.86%
12	0.0710	0.1389	97.96%	91.43%
13	0.0669	0.1363	97.75%	92.86%
14	0.0776	0.1452	97.55%	90.00%
15	0.0986	0.1469	96.73%	91.43%
16	0.0713	0.1462	97.96%	94.29%
17	0.0581	0.1417	98.16%	94.29%
18	0.0544	0.1412	98.57%	91.43%
19	0.0497	0.1488	98.36%	91.43%
20	0.0653	0.1340	97.55%	92.86%

QNN Noisy

Metric	Value
Accuracy	75.71%
Precision	75.06%
Recall	75.71%
F1 Score	75.12%
Loss	0.6106
Train Loss	0.5754
Val Loss	0.6106
Train Accuracy	79.55%
Val Accuracy	72.86%
Train Size	489
Val Size	70
Test Size	140

QNN Noisy Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	79	13
Class 1	21	27

QNN Noisy Classification Report

Class	Precision	Recall	F1 Score	Support
0	79.00%	85.87%	82.29%	92.0000
1	67.50%	56.25%	61.36%	48.0000
macro avg	73.25%	71.06%	71.83%	140.0000
weighted avg	75.06%	75.71%	75.12%	140.0000
Accuracy			75.71%	

QNN Noisy Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.7219	0.6891	29.24%	28.57%
2	0.6773	0.6598	32.31%	47.14%
3	0.6519	0.6426	57.06%	58.57%
4	0.6316	0.6313	72.60%	68.57%
5	0.6206	0.6238	80.16%	70.00%
6	0.6103	0.6195	80.16%	72.86%
7	0.6015	0.6156	79.75%	74.29%
8	0.5953	0.6138	79.75%	72.86%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.5901	0.6124	79.75%	74.29%
10	0.5876	0.6101	79.55%	74.29%
11	0.5836	0.6110	79.14%	72.86%
12	0.5796	0.6100	79.96%	74.29%
13	0.5799	0.6095	79.55%	72.86%
14	0.5741	0.6102	79.35%	72.86%
15	0.5739	0.6097	79.55%	72.86%
16	0.5741	0.6103	79.75%	72.86%
17	0.5754	0.6106	79.55%	72.86%

QNN Mitigated

Metric	Value
Accuracy	77.14%
Precision	76.67%
Recall	77.14%
F1 Score	76.00%
Loss	0.5674
Train Loss	0.5801
Val Loss	0.5674
Train Accuracy	80.57%
Val Accuracy	74.29%
Train Size	489
Val Size	70
Test Size	140

QNN Mitigated Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	83	9
Class 1	23	25

QNN Mitigated Classification Report

Class	Precision	Recall	F1 Score	Support
0	78.30%	90.22%	83.84%	92.0000
1	73.53%	52.08%	60.98%	48.0000
macro avg	75.92%	71.15%	72.41%	140.0000
weighted avg	76.67%	77.14%	76.00%	140.0000
Accuracy			77.14%	

QNN Mitigated Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6835	0.6273	70.76%	77.14%
2	0.6573	0.6119	73.42%	81.43%
3	0.6399	0.6017	75.87%	75.71%
4	0.6252	0.5939	77.91%	77.14%
5	0.6183	0.5877	78.32%	77.14%
6	0.6082	0.5823	78.94%	75.71%
7	0.6003	0.5806	79.55%	74.29%
8	0.5958	0.5761	79.55%	74.29%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.5910	0.5726	79.75%	74.29%
10	0.5853	0.5716	79.75%	74.29%
11	0.5830	0.5691	80.16%	74.29%
12	0.5801	0.5674	80.57%	74.29%